Charilaos Mylonas, Ph.D.

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About

I am a Senior AI engineer, with demonstrable experience in scientific computing and maintainable software development. I hold a M.Sc. in Computational Science and Engineering and a Ph.D. on Applied ML for predictive maintenance. Moreover, I have expertise in the Finite Element Method, as I was tasked with teaching it as part of my teaching assistant duties during my Ph.D., and I have implemented FEM and other numerical PDE solution techniques from scratch. Finally, I have extensive practical experience with Ansys and other multi-physics software.

Work Experience

Lead AI Engineer and Technical Consultant (Agentic AI/Full-Stack/DevOps) Aug 2025 – current
EPAM, Zurich

- Built an internal LangGraph evaluation and benchmarking library to systematically test agent reliability, reducing token consumption by 35% across revised agent factory templates for target use cases.
- Architected the core backend framework for LangGraph-based GenAI agents for portfolio optimization and insights (Python, FastAPI)
- Led integration of the agentic AI platform with two legacy systems and created a modular, reusable orchestrator agent template.
- Implemented CI/CD pipelines and configured internal IaC templates (Terraform, Ansible, Containerization, GitLab).

Senior Machine Learning Engineer (ended due to strategic branch closure) Feb 2025 – Jun 2025
Modulai, Basel

- Implemented algorithms for large-scale community detection in transaction graphs, computed graph embeddings across the client's full user base of 35M+ users.
- Extended the client's boosted decision tree-based fraud detection pipeline to automatically compute graph-derived features in a scalable manner, resulting in up to 7% reduction in false positive rate on test data.

Senior Consultant; promoted to Assistant Manager in Sep 2024 Feb 2022 – Jan 2025
Deloitte, Zurich

- Co-led early GenAI Incubator initiatives, designing proof-of-concept RAG-based LLM prototypes and contributing technical expertise across NLP, graph machine learning, and compliance analytics use cases.
- Contributed to business development for AI in energy trading by shaping technical pitch content and taking part in client pitch sessions.
- Delivered AI, analytics, and financial-risk engagements across banking, compliance, blockchain, and energy-market use cases, combining hands-on machine learning engineering with client-facing advisory work, requirements translation, and stakeholder management.
- Introduced software engineering best practices across client and internal engagements, improving version control, testing, project tracking, accountability, and maintainability.

Ph.D. Candidate/Research Assistant Sep 2016–Nov 2021
ETH Zurich

- Performed research on probabilistic ML for structural health monitoring and predictive maintenance of wind energy infrastructure (Python, TensorFlow, Graph Machine Learning, conditional VAEs).
- Implemented a message-passing GNN library (`https://github.com/mylonasc/tf-gnns/`).
- Performed large-scale Monte-Carlo simulations (Bash, cluster computing, typical volumes processed 200GB to 2TB)

Research Assistant Dec 2015–Sep 2016
ETH Zurich

- Contributed to the popular computational statistics software UQLab by implementing uncertainty quantification and sensitivity analysis algorithms.

Full-Stack Trading Tool Developer (internship) Jul 2014–Dec 2014
Credit Suisse, Zurich

- Implemented and validated a high-level interface for an option pricer (C++ to R), achieving more than 10-fold improvement by replacing pre-existing interface.

Education

Sept 2016 – Sept 2021	**ETH Zurich** Ph.D. in Machine Learning for Structural Health Monitoring under Uncertainty Advisor: Prof. Eleni Chatzi
Sept 2012 – Sept 2015	**ETH Zurich** M.Sc. in Computational Science and Engineering Specialization: Computational Electromagnetics Advisor: Prof. Ralf Hiptmair

Technical Strengths

Programming Languages	Python, Matlab, R	●●●●●●
	C++, Java, JavaScript	●●●●○○

Other software development skills Linux, Docker, Kubernetes, Classical ML Algorithms, Scientific Computing, Software Design, Web Development, High Performance Computing, Retrieval Augmented Generation systems, Microcontroller Programming

Deep Learning Probabilistic Generative Models (GANs, VAEs, Normalizing Flows, Denoising Diffusion models), Graph Neural Networks, Strong familiarity of all core Deep Learning architectures (gated RNNs, CNNs, Attention Mechanisms & Transformers) and how they apply to different data modalities (text, audio, images, tabular data).

Other Information

Teaching assistant roles

- High Performance Computing for CSE (C++, OpenMP) (2020) (Prof. O. Schenk).
- Method of Finite Elements (Matlab, Python) (2017 – 2019) (Prof. E. Chatzi).

Other academic engagements

- *Mentorship*: Serving as mentor for Ph.D. students at ETH Zurich (upon invitation).
- *Student project supervision*: 6 MSc theses and semester projects and consulted on several others.
- *Reviewer assignments*: for Mechanical Systems and Signal Processing and Journal of Sound and Vibration.

Distinctions and certificates

- **Best paper award** in 39th IMAC conference (Feb. 2021).
- **SIAM Gene Golub Scholarship** for Ph.D. summer school on "*High-Performance Data Analytics*" *Aussois, France 2019.*

Selected Publications

Please refer to Google Scholar [link] for full list and updated citation count.

Mylonas, C. *(ETH Ph.D. Dissertation)* Machine Learning for Structural Health Assessment under Uncertainty, with applications in Wind Energy, [link]

Mylonas C, *Chatzi E.* Remaining Useful Life Estimation for Engineered Systems Operating under Uncertainty with Causal GraphNets. Sensors. 2021; 21(19):6325. https://doi.org/10.3390/s21196325

Mylonas, C., *Abdallah, I., Chatzi, E.* Conditional variational autoencoders for probabilistic wind turbine blade fatigue estimation using SCADA data. Wind Energy. 2021; 1- 18. https://doi.org/10.1002/we.2621

Lai, Z., ***Mylonas, C.***, *Nagarajaiah, S., & Chatzi, E.* Structural identification with physics-informed neural ordinary differential equations. Journal of Sound and Vibration, 508, 116196.

Mylonas, C., *Abdallah, I., Chatzi, E.* (2021) Relational VAE: A Continuous Latent Variable Model for Graph Structured Data [link]

Mylonas, C., *Tsialiamanis, G., Worden, K. & Chatzi, E.* Bayesian graph neural networks for strain-based crack localization. (*39th IMAC conference proc.*) [link]

Mylonas, C., *Abdallah, I., & Chatzi, E. (2020).* Deep Unsupervised Learning For Condition Monitoring and Prediction of High Dimensional Data with Application on Windfarm SCADA Data. *In Model Validation and Uncertainty Quantification, Volume 3 (pp. 189-196). Springer, Cham.*